

Md. Emam Zafor Saadik

AI Engineer • ML Researcher • Web & Mobile Application Developer

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PROFESSIONAL SUMMARY

Final-year CSE student building research-driven, explainable AI systems across medical computer vision, Bangla NLP, and retrieval-augmented generation. Experienced in full-stack and embedded development, from diagnostic web platforms to edge-deployed assistive devices. Pursuing a career in academic research and teaching, focused on impactful, reproducible work.

TECHNICAL SKILLS

Frameworks: PyTorch, TensorFlow/Keras, Django, Flask, FastAPI, React, Bootstrap, Electron, Scikit-learn, Node.js, Express.js
Deep Learning: CNNs, Dual-Stream CNN Architecture, Transfer Learning, Metric Learning, Vision Transformers, Focal Loss, Image Augmentation
Computer Vision: Deepfake Detection, Medical Image Analysis, Face Detection & Alignment (MTCNN), Feature Extraction
Generative AI & Retrieval: LLMs/SLMs, RAG, Dense Retrieval, GANs, Agentic AI (Gemini, GPT-4o, Claude)
Programming Languages: Python, JavaScript, C, C++, PHP
Databases & Cloud: MySQL, MongoDB, Firebase/Firestore, ChromaDB
Embedded & IoT: Arduino, ESP32, ESP8266/NodeMCU, Hugging Face model deployment on edge hardware
MLOps & Tools: Google Colab, Git/GitHub, Linux (Kali, Ubuntu)
Mobile Development: Android
Graphics & Simulation: OpenGL, GLSL shaders, particle systems, physically based rendering

EDUCATION

Bangladesh University of Business & Technology (BUBT) Jan 2023 – 2027
B.Sc. in Computer Science & Engineering | Final-year student | CGPA: 3.85
Shahid Smrity Degree College 2018 – 2020
Higher Secondary Certificate (HSC) | GPA: 5.00
Government Swarupkati Pilot Model Secondary School 2013 – 2018
Secondary School Certificate (SSC) | GPA: 4.83

WORK EXPERIENCE

ARI Software Lab 2025 – Present
CTO & UI/Front-End Developer | Co-Founder
• Co-founded the software lab and lead UI/front-end development and technical direction across its published Android applications.
Peer Research Lab 2025 – Present
Research Intern
BUBT Photography Club 2023 – Present
Member

RESEARCH AND PUBLICATION

All works listed below are unpublished / currently under preparation.

TERRA: Temporal-Evolution and Reasoning-Trace RAG Python / LLMs / Dense Retrieval
• Built a dual-index RAG architecture combining an Event Evolution Graph for causal chains with a compressed reasoning-trace database, plus topological graph analytics and automated multi-phase LLM benchmark evaluation.
• Designed a dynamic retrieval router and an NLI-based semantic entailment grader, replacing keyword-matching evaluation to reduce temporal reasoning errors.
Lung Cancer Detection Using X-Ray Images React / FastAPI / MongoDB / PyTorch
• Engineered LungDx, a full-stack diagnostic platform with a DenseNet-169 inference engine and automatic PyTorch/TFLite fallback, classifying chest X-rays into 4 risk tiers (Low/Medium/High/Critical).
• Achieved sub-2-second inference latency via an async FastAPI + MongoDB backend, with secure auth, medical history tracking, and a doctor directory portal.

AQI Prediction – LSTM/BiLSTM vs. Classical ML Baselines

Python / TensorFlow

- Built a 5-layer AQI forecasting pipeline over 225,816 hourly Dhaka air-quality records, engineering 154 temporal/lag/interaction features and reducing them to 76 via variance, correlation, and multicollinearity filtering (50.3% reduction).
- Identified PM2.5 as the dominant AQI predictor (Pearson $r = 0.964$) and built a leakage-free chronological 70/15/15 split with SHAP-based interpretability to benchmark LSTM/BiLSTM against classical ML baselines.

GAN-BanglaFake: Automated Generation of Bengali Fake News Using GANs

Python / PyTorch / GANs

- Developed a conditional GAN generating synthetic Bangla news across 11 topic categories, trained on 1,299 hand-annotated and 4,812 crawled samples from 121 and 84 sources respectively.
- Improved a downstream BiLSTM fake-news classifier's accuracy from 93.1% to 97.4% (F1: 0.948 to 0.986) by augmenting training data with GAN-generated samples.

PROJECTS

ObsidianIDE – Cloud-Based Multi-Access Web IDE

React / Firebase / Node.js / Express

- Built a collaborative cloud IDE with Monaco editor integration, role-based access control, and a Git-style linear patch review system across 8 core views.
- Cut Firestore read costs by 90% via flat-path tree parsing and integrated a multi-model AI coding assistant (Gemini, GPT-4o, Claude) with one-click patch application; shipped with a 100% test-case pass rate across 24 test cases.

Blood Bank and Donation Management System

PHP / MySQL / Bootstrap

- Designed a 7-table relational schema and PHP/MySQL backend for donor registration, blood-type matching, and CMS-driven content across 8 blood groups.
- Built a randomized donor-matching engine (top-5 results per query) and a session-secured admin dashboard with full CRUD and query ticketing.

Smart Blind Stick with Bangladeshi Currency Detection

Arduino / ESP32 / Hugging Face

- Built an ESP32-based assistive device integrating ultrasonic, PIR motion, and GPS modules for real-time obstacle and environment mapping.
- Deployed an on-device Hugging Face vision model for real-time Bangladeshi currency recognition with audio feedback.

Cursion-2 – Real-Time Black Hole Lifecycle Simulator

OpenGL / GLSL / Team Project

- Contributed core physics logic (N-body Velocity Verlet integration, orbital mechanics, gravitational collapse behavior) to a 3-person real-time OpenGL/GLSL astrophysical simulator modeling an 18-phase cosmic lifecycle.
- Co-authored technical documentation covering the simulation architecture and physics methodology.

Stance Detection

Python / NLP

- Developed a multimodal NLP framework using Target-aware Graph Contrastive Learning for context-aware stance classification.
- Validated robustness on vaccine-related misinformation tasks across multimodal meme datasets, including TikStance.

RESEARCH INTERESTS

Medical Computer Vision • Deepfake Detection • Bangla NLP • Explainable Vision Models • Retrieval-Augmented Generation • Edge AI